Computational Investigation of a 16-Blade Axial Swirler for Enhanced Fuel–Air Mixing and NOx Mitigation in Non-Premixed Combustion Systems

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***Abstract*—The geometric configuration of a swirler is a primary determinant of flow dynamics, flame stabilization, and pollutant formation in non-premixed combustion chambers. This paper presents a focused computational study of a 16-blade axial swirler integrated into a cylindrical combustion chamber operating under non-premixed (diffusion) combustion conditions. Three-dimensional CFD simulations were conducted in ANSYS Fluent using the k–ω SST turbulence model and a non-premixed combustion formulation. The 16-blade configuration was characterized for velocity distribution, static pressure variation, temperature field, turbulence kinetic energy (TKE), nitrogen oxide (NOx) mass fraction, and soot formation across five axial planes—inlet, Zone 1, Zone 2, outlet, and cross-section. The results demonstrate that the 16-blade swirler generates the highest swirl intensity and TKE among the configurations studied, with an inlet TKE of 235.667 J/kg compared to 30.941 J/kg for the 8-blade design. Inlet velocity rises to 64.94 m/s and outlet pressure drops to the lowest recorded value of 30.54 Pa, confirming superior flow energy dissipation. However, the configuration exhibits the highest NOx mass fraction at the outlet (2.28×10^{-7}) and elevated soot levels (1.87×10^{-10}), attributable to intensified high-temperature recirculation zones. These findings characterize the performance trade-offs inherent in high-blade-count swirler designs and provide critical data for low-emission gas turbine combustor optimization.**

***Keywords*—**16-blade axial swirler, non-premixed combustion, CFD, ANSYS Fluent, central recirculation zone, turbulence kinetic energy, NOx emissions, soot formation, swirl intensity, flame stabilization, pressure drop, fuel–air mixing, gas turbine combustor.

I. Introduction

Efficient fuel–air mixing and stable flame anchoring are the two most essential requirements of a combustion system in gas turbine engines and industrial burners. Swirlers are one of the most powerful devices that serve two purposes simultaneously. An axial swirler adds a tangential velocity component to the incoming air stream creating a highly turbulent flow field and establishing a Central Recirculation Zone inside the combustor. The CRZ entrains hot combustion products back toward the injector face, which effectively provides a continuous ignition source that stabilizes the diffusion flame across loading conditions.

In most non-premixed combustion systems, fuel and oxidizer enter the chamber separately and depend solely on turbulent mixing before the reaction occurs. The mixing process is responsible for the combustion efficiency, peak flame temperature and generation of emissions like NOx and soot. Thus, the swirler geometry, which mainly relates to swirler number, has strong impact on most of the downstream combustion characteristics.

In this research, a systematic computer study based on ANSYS Fluent CFD software is performed for the 16-blade axial swirler configuration. This analysis contains the aerothermal and chemical characterization of this configuration through pressure, velocity, temperature, turbulence kinetic energy, mass fraction of NOx and mass fraction of soot contours for five axial planes.

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A. Role of Swirl in Non-Premixed Flames

Stabilization of the flame front against blow-off at sufficiently high flow velocities is one of the significant challenges in diffusion flames. Swirlers inject a high velocity tangential component into the incoming airstream and create a CRZ under the swirler exit plane. This CRZ provides continuous upstream transport of hot combustion products, which encourages ignition of the incoming fuel–air mixtures and supports stable lean combustion.

Swirl number S, the ratio of axial flux of angular momentum to the product of axial flux of linear momentum and a characteristic radius, quantifies swirl intensity. The best flame stability and pressure loss balance is achieved when non-premixed systems exhibit moderate swirl numbers (0.4–1.0). Higher counts of blades shift this balance more towards stronger swirl and turbulence, but also with the penalty of increased drag and loss of lift.

B. Importance of Blade Count in Diffusion Combustion

In contrast, for the base configuration of axial swirlers used in this work smaller number of blades produce a higher inter-blade passage width and allows to flow through larger channels. This speeds up the flow in each passage, enhances shear layer formation, and strengthens turbulence generation. The larger size of enhanced turbulence mixing zone surrounding the CRZ increases the loco motivity of fuel–air contact. But the smaller passages come at a cost in frictional and pressure losses, while at very high blade counts, flow separation between neighbouring blades can worsen mixing uniformity. It is important to learn this trade-off for swirler design.

II. Literature Survey

Khandelwal et al. [1] investigated axial swirler performance by varying vane number and angle in an annular gas turbine combustor using CFD, establishing pressure-loss and drag-coefficient based design procedures. Samiran et al. [2] evaluated a novel axial-radial combined swirler for non-premixed combustion using LPG fuel in ANSYS Fluent and demonstrated significant CO reduction compared to axial-only designs.

Zavaleta-Luna et al. [3] applied genetic algorithms to optimize swirl number and recirculation for a non-premixed combustor, producing longer recirculation zones and substantially reduced CO and NOx predictions. Lin et al. [4] numerically and experimentally studied a non-premixed double-swirl combustor and showed that increasing outer-swirler blade angle continuously reduces NO emissions. Ju et al. [5] compared flat-vane and curved-vane axial swirlers and found that curved vanes reduce pressure loss by 35% and NOx by 69%. Kumari [6] experimentally mapped the flow through a rotating 8-blade swirler and quantified how swirl number affects recirculation geometry.

A. Research Gap

Most of the current research focuses on isolated swirler cases or investigates one geometric parameter at the time. In the open literature, there are few systematic comparisons across blade counts—with complete characterization of velocity, pressure, temperature, TKE, NOx and soot—at the same non-premixed combustion boundary conditions. The gap this study fills is that it provides a complete set of data from CFD for the 16-blade configuration as part of a multi-configuration investigation.

B. Objectives

The study aims specifically to:(i) create a geometry of an axial swirler combustor with 16 blades using SolidWorks, and(ii) produce a mesh quality tetrahedral mesh containing 834,866 cells in ANSYS Meshing,(iii) perform a non-premixed combustion simulation using the k-ω SST turbulence model in ANSYS Fluent,(iv) extract and analyse pressure, velocity, temperature, TKE (Turbulent Kinetic Energy), NOx mass fraction contours at inlet, Zone1, Zone2 outlet and cross-section planes, and(v) assess the trade-off between increased mixing/TKE versus elevated NOx/pressure loss against respective results of 8 blade and 12 blade configurations.

III. Methodology

A systematic computational methodology was followed, comprising geometry modelling, mesh generation, solver configuration, simulation execution, and post-processing result analysis.

A. Geometry Modelling

The modelling of the 16-blade axial swirler is a cylindrical micro gas turbine combustion chamber in SolidWorks. The important geometric parameters considered are the outer swirler diameter, hub diameter, the blade thickness, blade spacing and number of blades (16 blades). The CAD geometry is an approximation of swirler configurations seen in real-world gas turbine combustors. Fig. Fig. 1 Cross-section view of the 16-blade design.

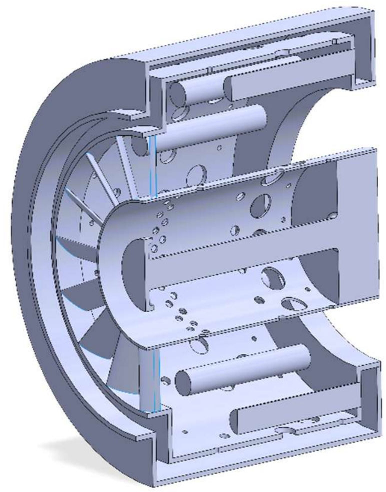

Fig. 1 CAD geometry of the 16-blade swirler with combustor (cross-sectional view).

B. Mesh Generation

The computational domain was discretized using ANSYS Meshing with tetrahedral cells. Mesh refinement was applied at the swirler blades, combustion chamber inlet, and recirculation zones to accurately resolve swirl motion, turbulence generation, and mixing behaviour. Fig. 2 shows the tetrahedral mesh, and Table I summarises mesh quality across the three configurations evaluated in the wider study.

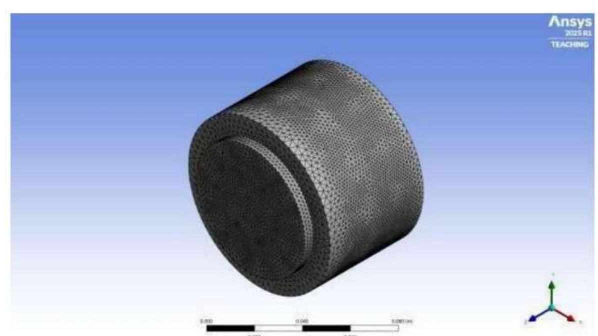

Fig. 2 Tetrahedral mesh of the 16-blade swirler–combustor assembly.

TABLE I
Mesh Size and Quality for All Configurations

Geometry	Cells	Faces	Nodes	Min. Orth. Quality
16 Blade	834,866	1,723,302	164,913	1.90×10^{-4}

C. Solver Setup and Boundary Conditions

All simulations were performed in ANSYS Fluent with a pressure-based solver. The k–ω SST turbulence model was selected for its accuracy in resolving swirling flows with adverse pressure gradients. The non-premixed combustion model was applied with the energy equation enabled. Table II summarises the simulation parameters.

TABLE II
CFD SIMULATION PARAMETERS

Parameter	Value
Solver Type	Pressure-Based
Turbulence Model	k–ω SST
Combustion Model	Non-Premixed
Energy Equation	Enabled
Air Inlet (Mass Flow)	0.49 kg/s @ 750 K; MMF = 1; TI = 5%
Fuel Inlet (Mass Flow)	0.02 kg/s @ 310 K; TI = 10%
Outlet	Gauge P = 0 Pa; Back-flow T = 300 K
Soot Model	Mass Brookes
NOx Model	Thermal NOx
Radiation	P1 Model
Iterations	1000 (first 100 @ 1st order, then 2nd order)

IV. RESULTS AND ANALYSIS – 16-BLADE AXIAL SWIRLER

CFD post-processing was carried out for six flow parameters across five axial planes (Inlet, Zone 1, Zone 2, Outlet, and Cross-Section). The following subsections present and discuss each parameter in detail.

A. Pressure Contour Analysis

For the 16-blade swirler, the range in pressure contours is roughly -1.84×10^4 Pa (dark blue) to $+4.99 \times 10^4$ Pa (red). Stress builds in the central hub region due to flow stagnation and rotational compaction by the 16 closely-packed blades. When fluid passes through the radial passage in between blades, pressure reduces in an orderly manner from yellow-green to light blue, reflecting the transfer of momentum from static pressure to dynamic pressure. Recirculation-driven pressure recovery is indicated by localised orange/red patches at the boundary of the outer wall.

1) Inlet Pressure Contour: The inlet plane shows a mostly orange centre with radial channels transitioning from yellow green to light blue outward. 16-blade inlet pressure was 13,674.6 Pa is the lowest consistent with flow resistance increasing with blades and statics pressure decreasing at entry).

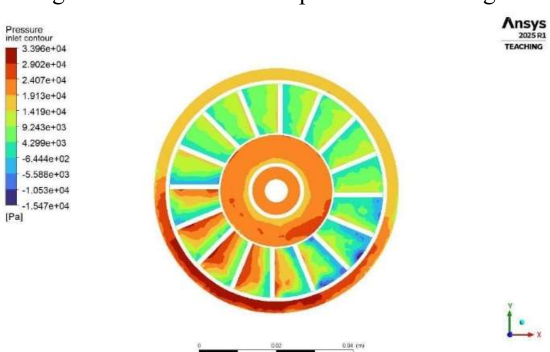

Fig. 3 Inlet pressure contour – 16-blade swirler.

2) Zone 1 Pressure Contour: Deep red at central Hub indicating high pressure with the annulus being teal-blue showing rapid pressure drop as flow expands radially. Compared to the 16-blade design, zone 1 pressure is improved of 9,063.71 Pa quantifying the bigger aerodynamic loss that increase number of blades impose.

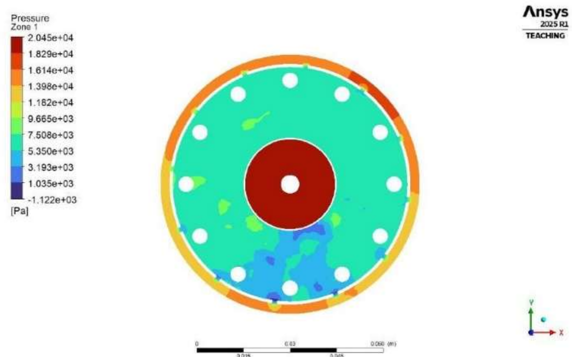

Fig. 4 Zone 1 pressure contour – 16-blade swirler.

*3) Zone 2 Pressure Contour :*Zone 2 shows the central high-pressure core surrounded by annular low pressure, represented by a blue ring at the outer boundary of which high-velocity outward jet flow occurs. The zone 2 pressure of 11,190.7 Pa demonstrates partial recovery compared to the zone 1 values consistent with flow reattachment effects.

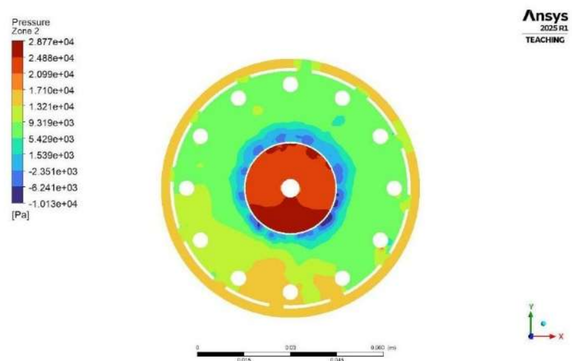

Fig. 5 Zone 2 pressure contour – 16-blade swirler.

4) Outlet Pressure Contour: The highest pressure near the inner circular boundary (red-orange) decreases outward to dark blue on the outlet annular plane. Outlet pressure = 30.54 Pa for all cases. The lowest pressure drop gives clues to the maximum energy dissipation done by the three-phase (16-blade) swirler.

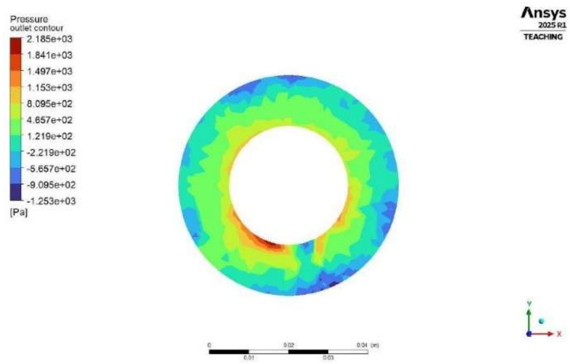

Fig. 6 Outlet pressure contour – 16-blade swirler.

5) Cross-Section Pressure Distribution: : View of the cross-section and the axial gradient (inlet (≈4.77×10^4Pa) to outlet (≈2.69×10^3Pa)) The total pressure drops accounts for the frictional and dynamic losses through 16 blade passages and the volume of combustor.

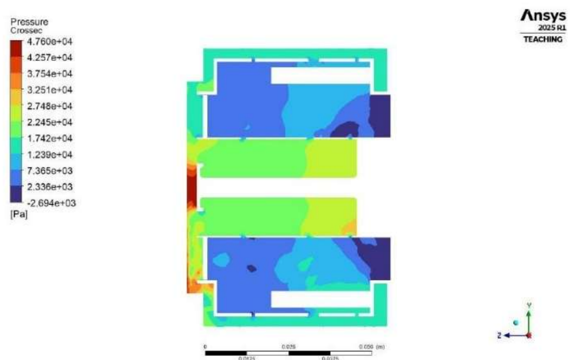

Fig. 7 Cross-section pressure contour – 16-blade swirler.

B. Velocity Contour Analysis

The contours of velocity for 16-blade configuration depicts that the maximum speed reaches almost 191.6 m/s in the areas of passing by vanes due to a strong flow acceleration taking place in inter-blade channels, where the blades are positioned in contact with each other. Due to hub blockage, the central core velocity (dark blue) remains nearly zero as flow disperses into the combustion chamber as moderate green-to-light-blue velocities in the outer annular region.

1) Inlet Velocity Contour: The 16-blade configuration produces a velocity at the inlet of 64.94 m/s, larger than the respective velocities, confirming that closer packing of blades accelerates flow towards their inlet.

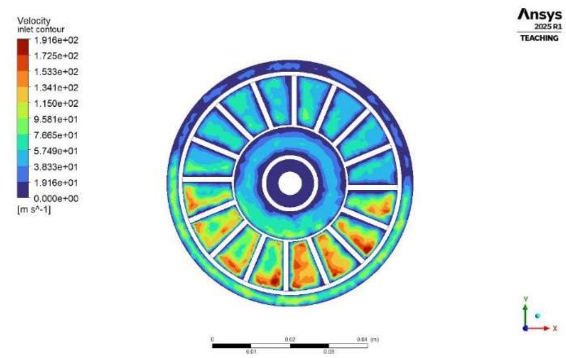

Fig. 8 Inlet velocity contour – 16-blade swirler.

2) Zone 1 Velocity Contour: Zone 1 is the evolution of swirling flow downstream of the swirler exit. Near the swirler, high-velocity patches are concentrated within the near-swirler mixing layer, and both these features appear in the central region as CRZ begins to develop. Velocity in zone 1 — 34.66 m/s (16-blade design)

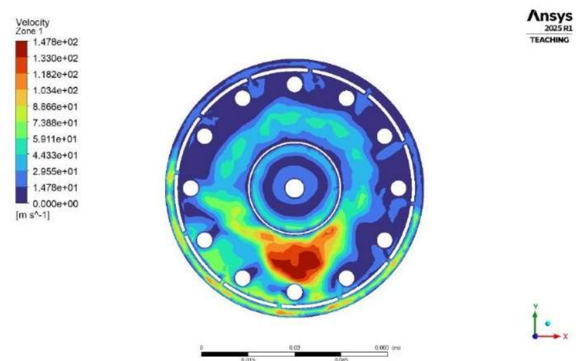

Fig. 9 Zone 1 velocity contour – 16-blade swirler.

3) Zone 2 Velocity Contour: Zone 2 velocity shows high-speed regions are more widely dispersed due to the swirling jet interacting with combustion products and walls of the chamber. The velocity in Zone 2 for the 16-blade configuration (V0 = 33.53 m/s) decelerates slightly from that of Zone 1, as expected due to energy transfer to turbulence.

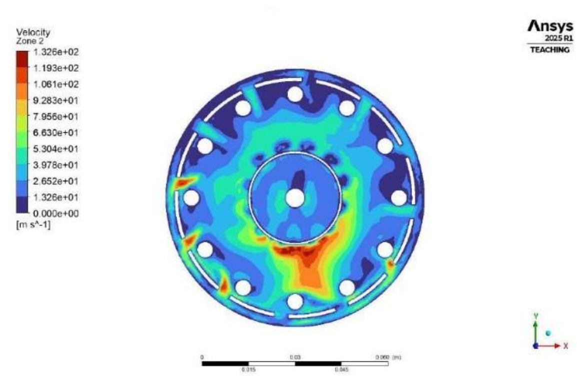

Fig. 10 Zone 2 velocity contour – 16-blade swirler.

4) Outlet Velocity Contour: The velocity distribution across the annular domain at the outlet appears moderate with some non-uniformity in the circumferential direction due to residual swirl. The outlet velocity of 59.71m/s for the 16-blade is much lower than the value of 96.35 m/s when there are eight blades, which indicates better energy dissipation by means of kinetic energy in this case.

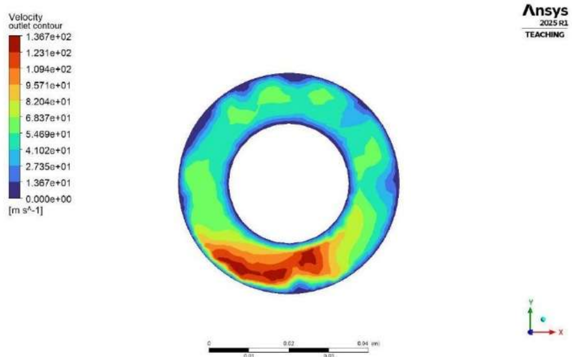

Fig. 11 Outlet velocity contour – 16-blade swirler.

5) Cross-Section Velocity and Vector Map: The cross-section view (Fig. 12) depicts the entire evolution of axial velocity. The swirler region contains the maximum velocities, decreasing behavior in the combustion zone, and then continuing this trend towards the outlet. The velocity vector map (Fig.13) clearly illustrates the rotational swirl structure and is validated with a well-organized central recirculation zone characteristic of high-swirl flows.

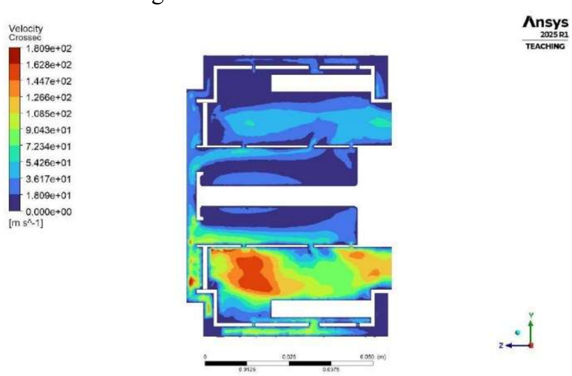

Fig. 12 Cross-section velocity contour – 16-blade swirler.

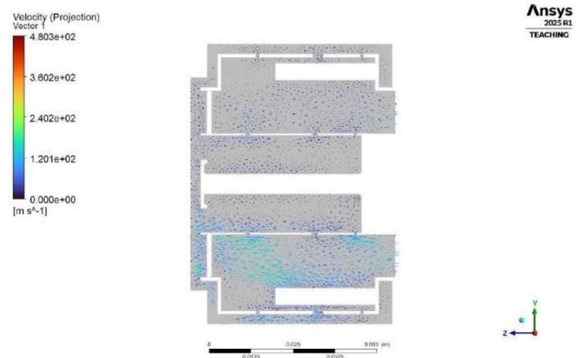

Fig. 13 Cross-section velocity vector map – 16-blade swirler showing central recirculation zone.

C. Temperature Contour Analysis

Temperature contours are drawn on the scale of 300 K (cool incoming air) to maximum combustion zone temperature levels around 8,200 K, respectively. Agitation intensity increases with increasing number of blades, resulting in a comparatively unique thermal distribution produced by the 16-blade swirler compared to the case of lower blade counts.

1) Inlet Temperature Distribution: The inlet temperature contour reveals cool vane passages in the upper half of the swirler, while moderate-to-warm temperatures occur in the lower half of the vane channels. The inlet mean temperature for the 16-blade configuration is found to be 1,026.69 K, well indicating that increasing airflow through the extra passages allows more inlet cooling and mixing to take place.

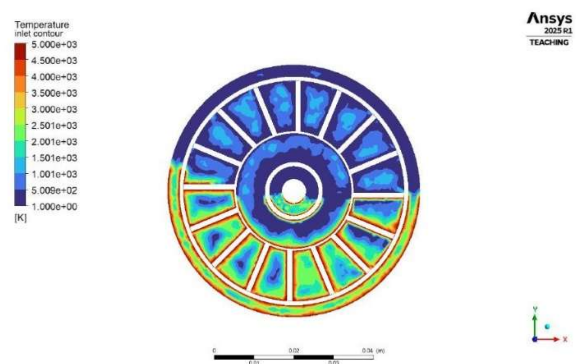

Fig. 14 Inlet temperature contour – 16-blade swirler.

2) Zone 1 Temperature Distribution: Temperature distribution in zone 1 is mainly low temperature with warm patches in the lower portion and orange-red bands near the inner hub along with the outer boundary. In this regard, the 906.77 K of Zone 1 temperature indicates a significant reduction. thus, avoiding overheating with increased airflow

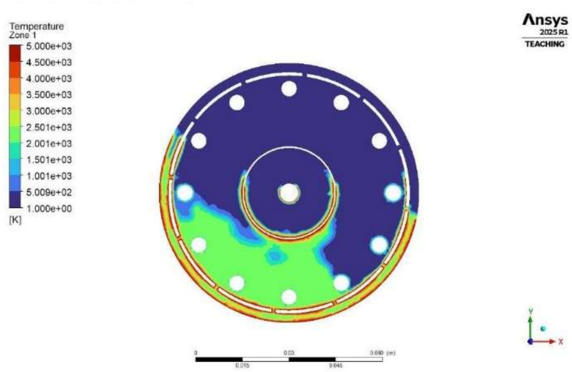

Fig. 15 Zone 1 temperature contour – 16-blade swirler.

3) Zone 2 Temperature Distribution: The thermal field in Zone 2 progresses further from that in both Zones 1 and the edge/core zone, with most of the adult green-to-yellow regions distributed around the combustion core. The combustion zone temperature of 1,031.51 K confirms that the 16-blade swirler keeps a lower-temperature combustion zone in Zone 2.

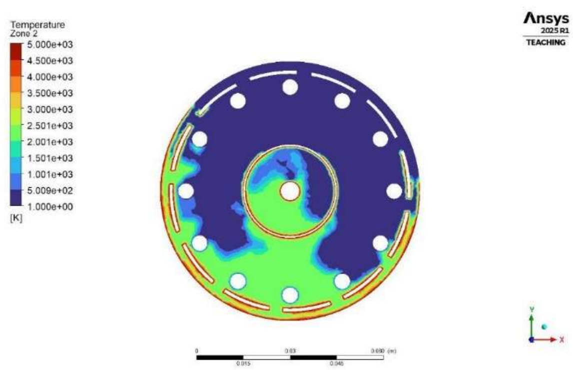

Fig. 16 Zone 2 temperature contour – 16-blade swirler.

4) Outlet Temperature Distribution: The outlet temperature for the 16-blade case was found to be 759.23 K. This major reduction point indicates that improved mixing promotes more complete combustion in the primary zone, cooler exhaust temperature delivered to the turbine blades, which has a significant effect on turbine blade thermal loading.

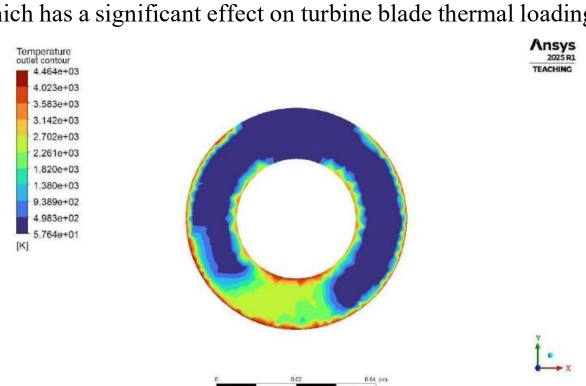

Fig. 17 Outlet temperature contour – 16-blade swirler.

5) Cross-Section Temperature Distribution: The cross-section snapshot affirms a sequential thermal progression from the hot supply inlet region to the cooled outlet section. As indicated by the symmetric upper–lower pattern, a balance of flow through both combustor passages is achieved.

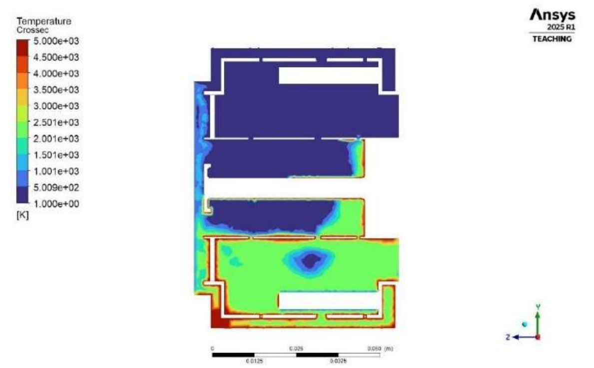

Fig. 18 Cross-section temperature contour – 16-blade swirler.

D. Turbulence Kinetic Energy (TKE) Analysis

The turbulence kinetic energy outcomes show the greatest difference of the 16-blade configuration, with inlet TKE amounts of 235.667 J/kg, about 7.6 times higher. The powerful wave and turbulence generation capacity of 16 nearly-touching blades passages is well seen in the illustrated strong oscillations from the average value.

1) Inlet TKE: The inlet TKE contour indicates asymmetric distribution in upper and lower passes of the vanes. Large TKE patches at interceptions of the vane walls indicating shear-layer driven turbulence production.

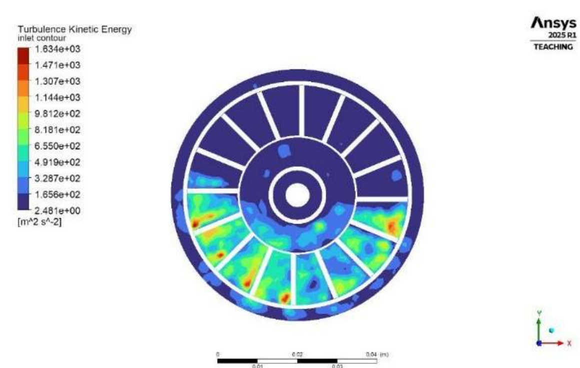

Fig. 19 Inlet TKE contour – 16-blade swirler.

2) Zone 1 TKE: For the case of the 16-blade configuration, the TKE calculated in Zone 1 is found to be equal to (154.164 J/kg), as seen from this figure, which indicates that a region of extremely high turbulence generated by increased blade number persists through the first combustion zone.

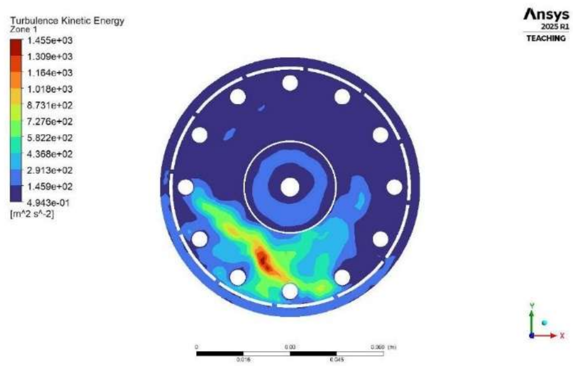

Fig. 20 Zone 1 TKE contour – 16-blade swirler.

3) Zone 2 TKE: The TKE of 152.632 J/kg that persists in this zone indicates turbulent mixing in the primary combustion zone is still present and high, (close to Zone 1 values). This enduring turbulence is effectively the main reason that the 16-blade swirler provides better fuel–air mixing.

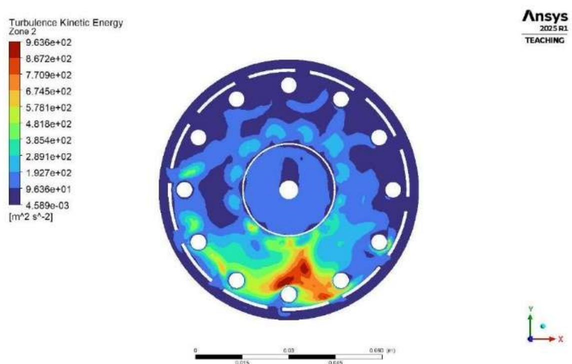

Fig. 21 Zone 2 TKE contour – 16-blade swirler.

4) Outlet TKE: The outlet TKE (109.395 J/kg) is lower than the combustion zone, which agrees with turbulence energy dissipation as the flow expands toward the exit. The controlled turbulence reduction at the outlet is an attractive feature for downstream turbine components.

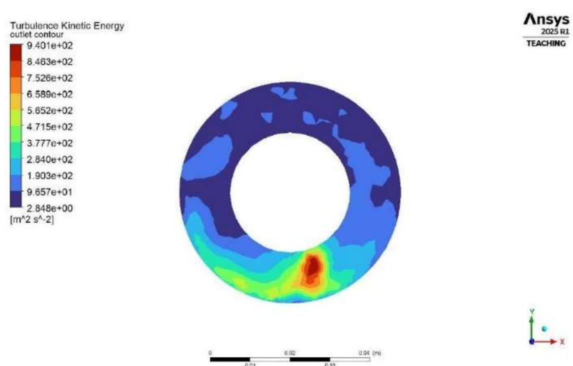

Fig. 22 Outlet TKE contour – 16-blade swirler.

5) Cross-Section TKE: The view for the cross-section component of the turbulent kinetic energy (TKE) shows largest turbulence localized at the swirler and near-swirler zone, then gradually diminishing towards the outlet. The present analysis affirms the 16-blade swirler as the main turbulence producer.

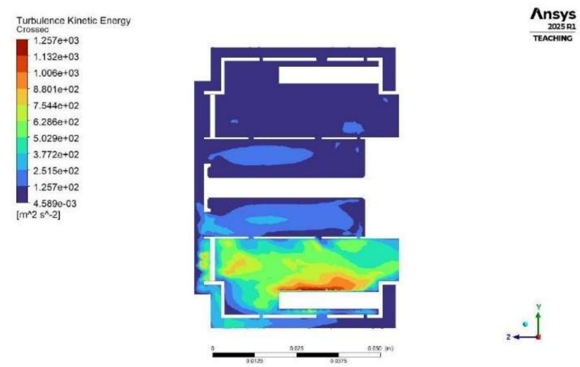

Fig. 23 Cross-section TKE contour – 16-blade swirler.

E. NOx Mass Fraction Analysis

A thermal NOx formation model was modified and used to predict nitrogen oxides generated. NOx formation takes place primarily in regions of high temperature and oxygen availability, being heavily dependent on the peak flame temperature. However, for NOx levels, the 16-blade configuration has the highest values of all three configurations (details on this are presented in sections) due to its intensified recirculation and local high-temperature zones close to the centre of combustion.

1) Inlet NOx: Inlet NOx for the 16-blade configuration is 6.02×10^{-11}, substantially higher in the inlet section as the air enters through the swirler, and high turbulence is created in the combustion chamber due to the mixing of air and fuel.

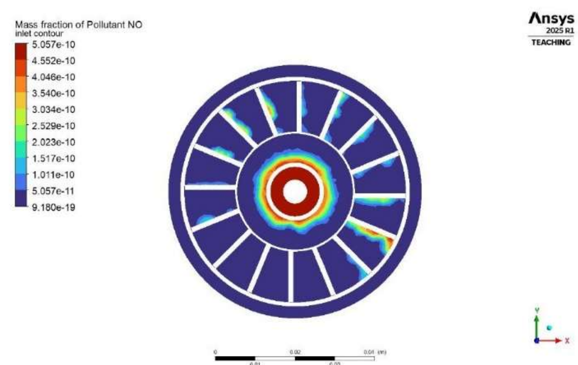

Fig. 24 Inlet NOx mass fraction contour – 16-blade swirler.

2) Zone 1 NOx: Since combustion is most intense at the left outer-wall region, Zone 1 has the maximum NOx concentration. Zone 1 NOx is 3.53×10^{10}.

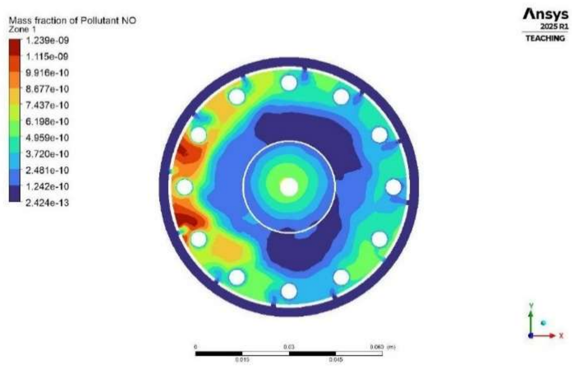

Fig. 25 Zone 1 NOx mass fraction contour – 16-blade swirler.

3) Zone 2 NOx: Characteristics of localised NO formation in high-temperature regions near the outer wall borderline of each zone at these concentrations. For the 16-blade design, Zone 2 NOx of 5.65×10^{-10} is the maximum spatial NOx accumulation.

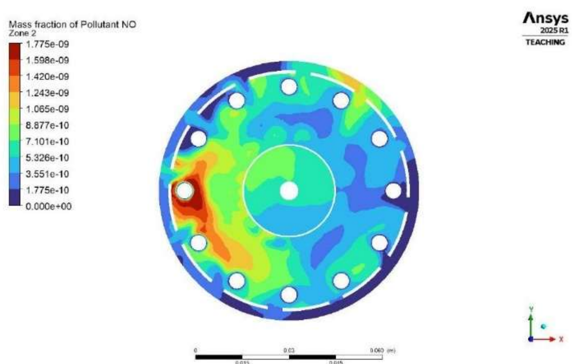

Fig. 26 Zone 2 NOx mass fraction contour – 16-blade swirler.

4) Outlet NOx: Outlet NOx of 2.28×10^{-7} for the 16-blade configuration is the highest, further confirming that intensified recirculation as well as persistent high-temperature zones facilitate enhanced formation of NOx

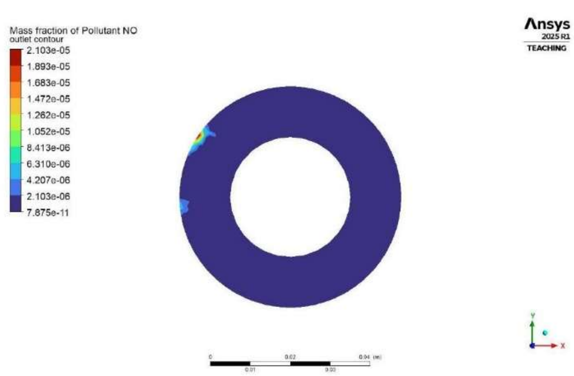

Fig. 27 Outlet NOx mass fraction contour – 16-blade swirler.

5) Cross-Section NOx: Cross-sectional NOx distributions indicate that very little NO makes it into the interior of the combustor A higher level of NOx emerges in the upper-right outlet region, where low-speed flowing combustion products linger, and temperatures are highest.

Fig. 28 Cross-section NOx mass fraction contour – 16-blade swirler.

F. Soot Mass Fraction Analysis

In non-premixed combustion, soot-forming conditions are locally attached to the fuel-rich zones characterized by partially burned reactants. Soot transport model: Mass Brookes soot model. Although absolute soot concentration levels are low for all cases investigated here, the 16-blade design has a certain spatial variation of soot distribution.

1) Inlet and Zone 1 Soot: Inlet soot for the 16-blade configuration is 2.04×10^{-13} and Zone 1 soot is at its maximum in both geometries at a value of 5.94×10^{-13}, with correspondingly lower values of pour throughout the injector compared to those loaded with the less-swirled jet, which would ultimately represent a concentration gradient higher than that expected using the same methodology for the combusted fuel mass classification in these near-injector regions under high-swirl conditions.

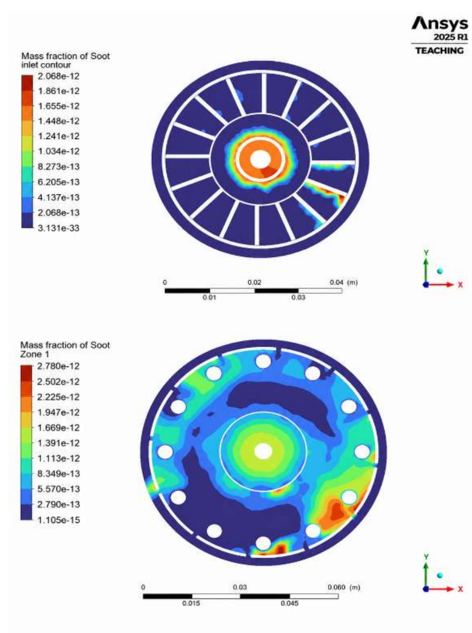

Fig. 29 Inlet (top) and Zone 1 (bottom) soot mass fraction contours – 16-blade swirler.

2) Zone 2, Outlet and Cross-Section Soot: In zone 2 soot (1.25 × 1010), soot particles spreading to a wide area compared in general, soot levels are again very low throughout most of the domain at the outlet (1.87 × 10−10), ultimately demonstrating that a majority of soot created in the primary zone erodes before entering the exhaust.

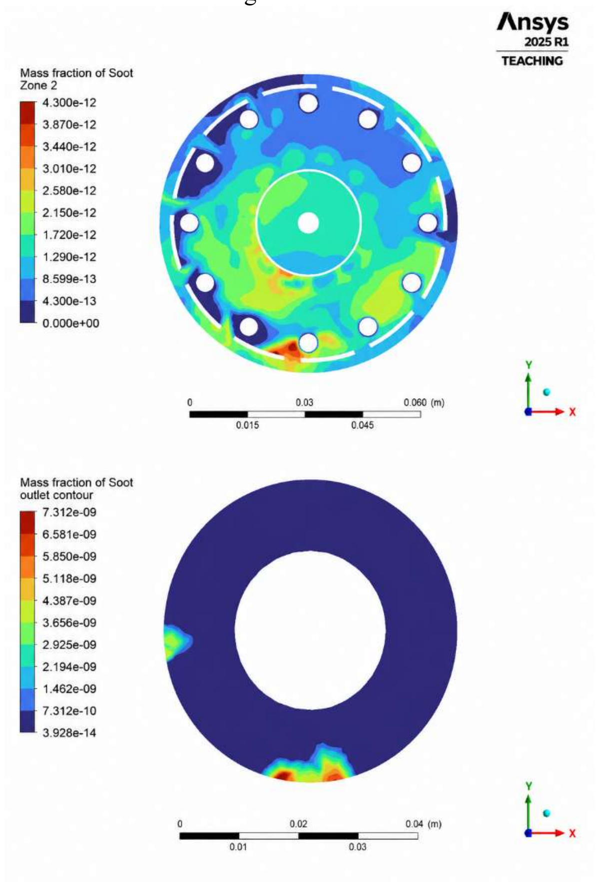

Fig. 30 Zone 2 (top) and outlet (bottom) soot mass fraction contours – 16-blade swirler.

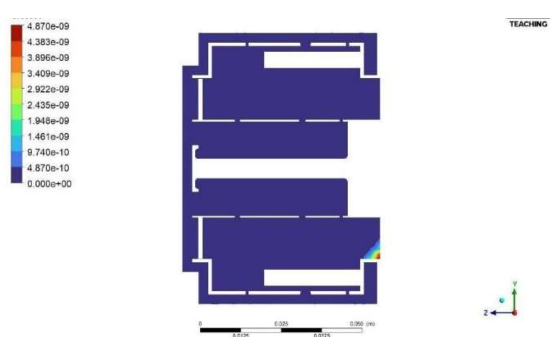

Fig. 31 Cross-section soot mass fraction contour – 16-blade swirler.

V. Performance Graphs of 16-Blade Configurations

For the purpose of context, all relevant parameters at similar boundary conditions are shown in 16-blade performance. The graphs below show the full data set for velocity, pressure, temperature, TKE, NOx mass fraction, and soot mass fraction plotted over four axial planes: Inlet, Zone 1, Zone 2, and Outlet.

A. Velocity vs. Flow Regions

TABLE III
Velocity Values (m/s) Across Zones

No. of Vane	Inlet	Zone 1	Zone 2	Outlet
16	64.94	34.66	33.53	59.71

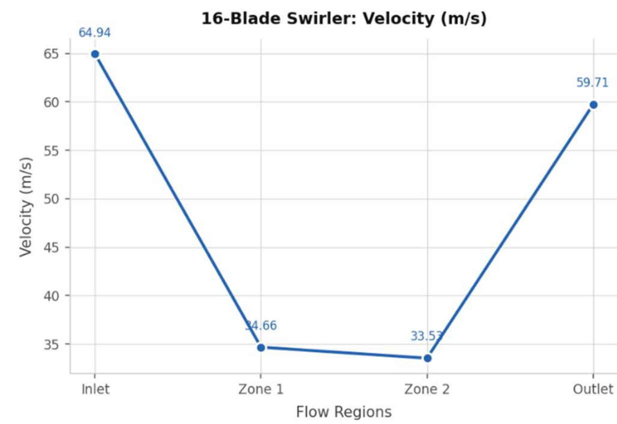

Fig. 32 Velocity vs Flow Regions.

This is achieved as incident air is accelerated by 16 narrow blade channels to a maximum inlet velocity of 64.94 m/s. It falls rapidly to 34 m/s in Zones 1 and 2 as jets begin to expand, transferring bulk kinetic energy into turbulence before increasing again to recover at the exit plane up to 59.71 m/s — far from the maximum eight-blade value of 96.35m/s, thus confirming superior energy dissipation.

B. Pressure vs. Flow Regions

TABLE IV
Pressure Values (Pa) Across Zones

No. of Vane	Inlet	Zone 1	Zone 2	Outlet
16	13,674.6	9,063.71	11,190.7	30.54

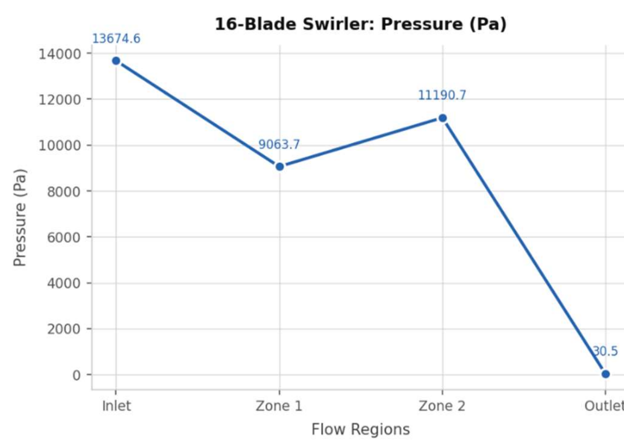

Fig. 33 Pressure vs. Flow Regions.

Pressure: drops from 13,675 Pa (inlet) to a near-zero-performing drop of 30.54 Pa (outlet) — steepest of any configuration. For Zone 1, records the minimum (9,064 Pa); moderate recovery in Zone 2 [11,191 Pa] corresponds to wall reattachment. This large mass loss is the aerodynamic penalty of 16 folders.

C. Temperature vs. Flow Regions

TABLE V
TEMPERATURE VALUES (K) ACROSS ZONES

No. of Vane	Inlet	Zone 1	Zone 2	Outlet
16	1,026.69	906.77	1,031.51	759.23

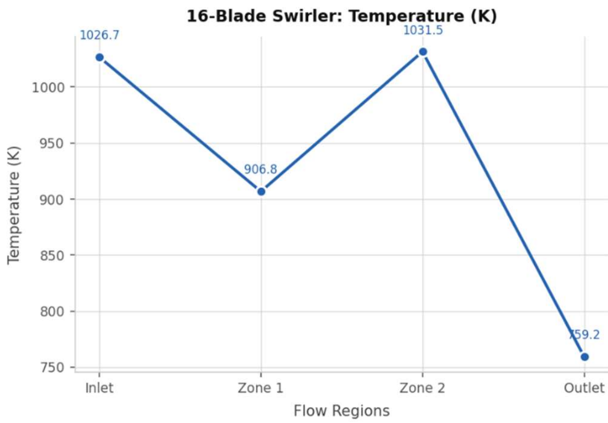

Fig. 34 Temperature vs. Flow Regions.

The 16-blade structure delivers the coolest temperatures in each area. The inlet indicates 1,027 K — 47% cooler than the 8-blade — due to higher dilution airflow. Zone 1 drops to reach 907 K, zone 2 slightly peaks up to mild temperatures at 1,032 K, and the outlet temperature is yielded at around 759 K, which will add thermal life for turbine blades.

D. Turbulence Kinetic Energy vs. Flow Regions

TABLE VI
TKE VALUES (J/KG) ACROSS ZONES

No. of Vane	Inlet	Zone 1	Zone 2	Outlet
16	235.667	154.164	152.632	109.395

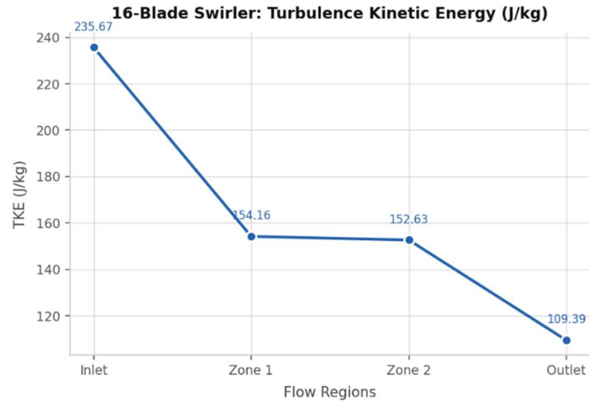

Fig. 35 TKE vs. Flow Regions.

Inlet TKE of 235.7 J/kg maximally exceeds the 8-blade driven by extreme shear at 16 blade trailing edges (30.9 J/kg) over a factor of 7.6× higher than the boundary layer level at the inlet edge, and up to this turbulence intensity is large enough that initial turbines cannot turn nor extract power from it. In contrast to lower blade counts where TKE increases downstream, here it decays in sequence — 154.2 (Zone 1) → 152.6 (Zone 2) → 109.4 J/kg (outlet), confirming effective utilization of turbulence for fuel–air mixing.

E. NOx vs. Flow Regions

TABLE VII
NOX MASS FRACTION ACROSS ZONES

No. of Vane	Inlet	Zone 1	Zone 2	Outlet
16	6.02×10^{-11}	3.53×10^{-10}	5.65×10^{-10}	2.28×10^{-7}

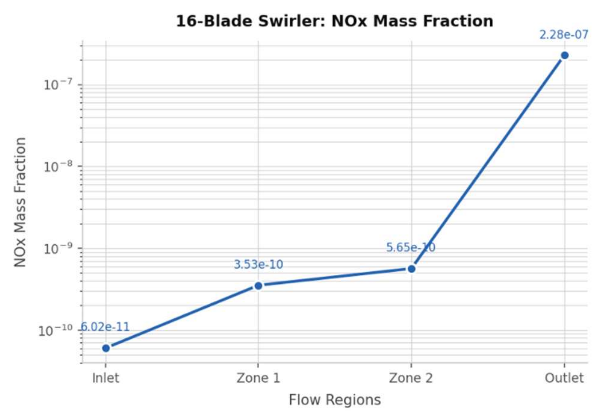

Fig. 36 NOx mass fraction vs. Flow Regions.

NOx is low in the internal zones but reaches $2.28\times10{-}7$ at the outlet. The Zeldovich thermal NOx mechanism, which is the main emission trade-off of this configuration, is driven by localised high-temperature pockets very close to the injector core.

F. Soot vs. Flow Regions

TABLE VIII
SOOT MASS FRACTION ACROSS ZONES

No. of Vanes	Inlet	Zone 1	Zone 2	Outlet
16	2.04×10^{-13}	5.94×10^{-13}	1.25×10^{-12}	1.87×10^{-10}

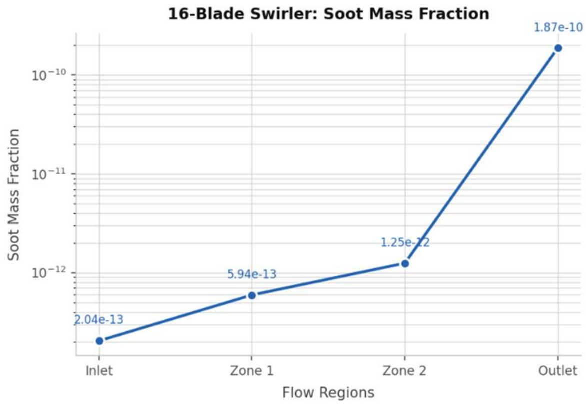

Fig. 37 Soot mass fraction vs. Flow Regions.

Soot builds monotonically from 2.04×10−13 (inlet) to 1.87×10−10 (outlet), as pockets fuel-rich in zones nucleate particles. The absolute values are also very low, suggesting efficient oxidation in the chamber. This suggests that while the 16-blade swirler improves turbulent mixing, the disrupted optimal flow pattern at very high blade counts allows some fuel-rich pockets to persist, leading to increased soot nucleation. All values remain extremely low in absolute terms.

VI. CONCLUSION

The present work has provided a detailed computational examination of the 16-blade axial swirler geometry designed for non-premixed combustion. The main findings of the ANSYS Fluent CFD investigation are:

(i) Higher turbulence kinetic energy (TKE); The 16-blade swirler provides the maximum TKE (inlet TKE =235.667 J/kg, around 7.6 × higher than that of the 8-blade configuration), which confirms that a swirler with a higher blade count is an efficient option to create turbulence for fuel–air mixing;

(ii) We also observed a significant variation in inlet velocity, increasing up to 64.94 m/s, while outlet velocity reduced to 59.71 m/s, showing more effective kinetic energy dissipation and almost uniform flow distribution at the combustible exit.

(iii) The overall pressure drop is the highest in 16 blades design, with inlet and outlet pressures of 13,674.6 Pa. and 30.54 Pa., respectively, which is the main aerodynamic drawback.

(iv) The NOx mass fraction at outlet (2.28×10^{-7}.) is the largest compared to the other two configurations due to localised high-temperature zones around the injector core where the Zeldovich thermal NOx mechanism occurs, although average temperatures are lower; and

(v) Soot formation is medium for 16-blade design with moderate soot levels at outlet: (1.87×10^{-10}). In general, the 16-blade swirler is a high-performance configuration for applications where vigorous fuel–air mixing and turbulence levels are most important. It must overcome its NOx penalty and pressure drop, which can be mitigated through geometric optimisation (blade angle, hub ratio) or staged combustion strategies before it can be applied in emission-constrained systems.

VII. FUTURE SCOPE

Directions for future research that appear productive are: 16-blade CFD results would benefit from validation through experimental work using particle image velocimetry (PIV) and laser-induced fluorescence (LIF); blade angle and the hub-to-tip ratio can be optimised to minimise NOx emissions whilst maintaining high TKE and mixing quality; further mitigating thermal NOx may be realised via staging combustion or employing lean premixed pre-evaporated (LPP) strategies; Large Eddy Simulation (LES) modelling may enhance resolution of unsteady coherent structures in the central recirculation zone; exhaustive multi-objective optimisation with genetic algorithms may reveal Pareto-optimal swirler geometries enabling simultaneous reductions in NOx-, soot- and pressure loss emissions [20]; finally, testing of the 16-blade geometry with alternative fuels including hydrogen blends and sustainable aviation fuels (SAF) will quantify how such operating regimes promote system flexibility under decarbonisation pathways.

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REFERENCES

[1] B. Khandelwal, D. Lili, and V. Sethi, “Design and study on performance of axial swirler for annular combustor by changing different design parameters,” J. Energy Inst., vol. 87, no. 4, pp. 372–382, 2014.

[2] N. A. Samiran, K. F. Abu Mansor, R. Abd Rashid, and M. S. S. Hamid, “CFD simulation analysis of non-premixed combustion using a novel axial-radial combined swirler for emission reduction enhancement,” CFD Lett., vol. 17, no. 7, pp. 18–31, 2023.

[3] D. A. Zavaleta-Luna et al., “Optimized design of a swirler for a combustion chamber of non-premixed flame using genetic algorithms,” Energies, vol. 13, no. 9, Art. no. 2240, 2020.

[4] J. Lin, M. Bao, F. Zhang, Y. Zhang, and J. Yang, “Numerical and experimental investigation of a non-premixed double swirl combustor,” Energies, vol. 15, no. 2, Art. no. 458, 2022.

[5] H.-J. Ju, J. H. Cho, and J. Hwang, “Numerical analysis of flow characteristics of a swirl-stabilized premixed burner with different swirl vane configurations,” J. Mech. Sci. Technol., vol. 36, pp. 5793–5800, 2022.

[6] M. Kumari, "Experimental analysis of flow through rotating combustion swirler with zero-degree inlet and outlet angle of guide vane," IJCSET, vol. 5, no. 8, pp. 820–824, 2014.

[7] K. Gupta, D. G. Lilley, and N. Syred, Swirl Flows. New York, NY, USA: Springer, 1984.

[8] A. H. Lefebvre and D. R. Ballal, Gas Turbine Combustion: Alternative Fuels and Emissions, 3rd ed. Boca Raton, FL, USA: CRC Press, 2010.

[9] T. Poinsot and D. Veynante, Theoretical and Numerical Combustion, 3rd ed. Edwards Publishing, 2011.

[10] V. N. Reddy, A. K. Agarwal, and S. K. Aggarwal, "Effect of swirler design on flame stabilization and emissions in gas turbine combustors," Fuel, vol. 238, pp. 312–323, 2019.